

a multiple regression. Three simple regressions, one after another, and the number will come out identical to the last decimal place.

In Chapter 12, we showed how to test hypotheses about individual coefficients and build confidence intervals around them. But what exactly does the coefficient on X measure when other variables are in the model? If I add a control variable to my regression, how does that change the coefficient on X , and why? The Frisch Waugh Lovell theorem answers these questions.

One of the most powerful results in econometrics is the **Frisch Waugh Lovell (FWL) theorem**. It answers a question that every applied researcher faces: when I run a multiple regression and look at the coefficient on one variable, what exactly is that coefficient measuring? The FWL theorem gives a concrete, step by step procedure that produces the same answer as the full regression, and in doing so, it reveals what “controlling for other variables” really means.

Here is the punchline up front: the coefficient on X in a regression of Y on X and Z is the same coefficient you get from a simple regression of the “cleaned” Y on the “cleaned” X , where “cleaning” means removing the influence of Z from both variables. Let’s make this precise.

13.1 Statement of the Theorem

Consider a full linear model with an outcome Y , a variable of interest X , and a set of control variables Z (which could be multiple explanatory variables):

$$Y_i = \beta_1 X_i + \gamma' Z_i + u_i,$$

for $i = 1, \dots, n$. (We absorb the intercept into Z for cleaner notation.) The FWL theorem states that the estimated coefficient $\hat{\beta}_1$ from the full regression can be obtained in three steps:

1. **Partial Z out of X .** Regress X on Z and save the residuals:

$$r = M_Z X, \quad \text{where } M_Z = I_n - Z(Z'Z)^{-1}Z'.$$

This is the residual maker matrix from Chapter 8, built for the controls Z instead of the full X . Just as $I_n - H$ produces the residuals from regressing on all of X , the matrix M_Z produces the residuals from regressing on Z alone. The residuals r are the part of X that cannot be explained by Z . Think of r as the “unique variation” in X .

2. **Partial Z out of Y .** Regress Y on Z and save the residuals:

$$v = M_Z Y.$$

These are the part of Y that cannot be explained by Z .

3. **Regress v on r .** Run a simple OLS regression of v on r . The slope coefficient from this regression is exactly $\hat{\beta}_1$.

In short: *controlling for Z is the same as removing Z from both sides and then running a simple regression.*

13.2 The Full Proof

The theorem claims that *one* regression can be replaced by *three*, and the suspicious part of that claim is where the extra regressions come from. We are going to run a single multiple regression through some algebra, and at a very specific moment two separate residual calculations are going to appear out of nowhere, one for X and one for Y . That moment is Step 6. Everything before it is bookkeeping, and everything after it is interpretation. If you only slow down once in this chapter, slow down there.

In Chapter 8 we derived the normal equations for a regression of $\mathbf{y}$ on a design matrix X , and got Equation (8.2):

$$X'X\hat{\beta} = X'y.$$

That is the whole toolkit. Everything below is that one equation, with the columns of the design matrix sorted into two piles.

Step 1: Sort the columns into two piles.

Our design matrix holds the variable we care about alongside the controls we are willing to tolerate. So let's physically separate them. Write the design matrix as D , with the X column first and the q columns of Z after it:

$$D = \begin{pmatrix} x_1 & z_{11} & z_{12} & \cdots & z_{1q} \\ x_2 & z_{21} & z_{22} & \cdots & z_{2q} \\ \vdots & \vdots & \vdots & & \vdots \\ x_n & z_{n1} & z_{n2} & \cdots & z_{nq} \end{pmatrix} = \left(\underbrace{X}_{n \times 1} \mid \underbrace{Z}_{n \times q} \right).$$

Nothing has happened yet. We have not deleted a column or changed a number. We have only drawn a vertical line down the middle of a matrix we already had, and given the two sides names. The controls Z include the intercept column of ones, which is why the theorem's statement said we absorb the intercept into Z : the intercept is a control like any other.

The coefficient vector splits the same way, because coefficients line up with columns:

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\gamma} \end{pmatrix}.$$

What is $\hat{\gamma}$? It is the $q \times 1$ vector of estimated coefficients on the control variables: one number for each column of Z , exactly the numbers your software prints in the rows below the variable you actually care about. We are never going to interpret them. But we cannot simply ignore them either, because $\hat{\beta}_1$ and $\hat{\gamma}$ are determined together, by the same system of equations. The entire proof is a campaign to make $\hat{\gamma}$ disappear.²

Step 2: Watch the normal equations split.

Now feed the partitioned D into $D'D\hat{\beta} = D'y$ and let the block multiplication happen. Stacking D' on top of itself:

$$D'D = \begin{pmatrix} X' \\ Z' \end{pmatrix} (X \ Z) = \begin{pmatrix} X'X & X'Z \\ Z'X & Z'Z \end{pmatrix}.$$

Each block is just the corresponding pair of pieces multiplied together, exactly as if X and Z were scalars. It is worth checking the sizes so the shapes stay honest:

$$\begin{pmatrix} \begin{matrix} X'X & X'Z \\ (1 \times 1) & (1 \times q) \end{matrix} \\ \begin{matrix} Z'X & Z'Z \\ (q \times 1) & (q \times q) \end{matrix} \end{pmatrix} \quad \text{which is } (1+q) \times (1+q) \text{ overall,}$$

and $(1+q)$ is the total number of columns in D , which is what $D'D$ should be. The right hand side splits the same way:

$$D'y = \begin{pmatrix} X' \\ Z' \end{pmatrix} Y = \begin{pmatrix} X'Y \\ Z'Y \end{pmatrix}, \quad \begin{matrix} X'Y, & Z'Y. \\ (1 \times 1) & (q \times 1) \end{matrix}$$

Put the three pieces together and the familiar normal equations become a partitioned system:

$$\begin{pmatrix} X'X & X'Z \\ Z'X & Z'Z \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} X'Y \\ Z'Y \end{pmatrix}.$$

This is still $X'X\hat{\beta} = X'y$. Same equation, same solution, same numbers. We have just organized it so that the variable we care about sits in its own row.

Step 3: Write out the two equations.

² Verbally, “gamma hat.” Bold because it is a vector, just like $\hat{\beta}$. If you prefer, imagine Z holds an intercept and three controls, so $\hat{\gamma}$ is four numbers stacked in a column.

$$X'X\hat{\beta}_1 + X'Z\hat{\gamma} = X'Y \quad (13.1)$$

$$Z'X\hat{\beta}_1 + Z'Z\hat{\gamma} = Z'Y \quad (13.2)$$

The top equation says: the X column's demands on the fit are met by $\hat{\beta}_1$ and $\hat{\gamma}$ working together. The bottom equation says the same thing for the Z columns. The two coefficients are tangled, and untangling them is the job.

Step 4: Solve (13.2) for $\hat{\gamma}$.

$$\begin{aligned} Z'Z\hat{\gamma} &= Z'Y - Z'X\hat{\beta}_1 && \text{Rearrange (13.2)} \\ \hat{\gamma} &= (Z'Z)^{-1}Z'Y - (Z'Z)^{-1}Z'X\hat{\beta}_1 && \text{Pre multiply by } (Z'Z)^{-1} \\ \hat{\gamma} &= (Z'Z)^{-1}Z'(Y - X\hat{\beta}_1) && \text{Factor out} \end{aligned}$$

Notice what that bought us. We now have $\hat{\gamma}$ written entirely in terms of things we know and one thing we want, $\hat{\beta}_1$. That is the lever. We can substitute this expression wherever $\hat{\gamma}$ appears, and the control coefficients will vanish from the problem.

Step 5: Substitute into (13.1).

$$\begin{aligned} X'X\hat{\beta}_1 + X'Z(Z'Z)^{-1}Z'(Y - X\hat{\beta}_1) &= X'Y && \text{Plug in } \hat{\gamma} \\ X'X\hat{\beta}_1 + X'H_ZY - X'H_ZX\hat{\beta}_1 &= X'Y && \text{Let } H_Z = Z(Z'Z)^{-1}Z' \\ (X'X - X'H_ZX)\hat{\beta}_1 &= X'Y - X'H_ZY && \text{Collect } \hat{\beta}_1 \text{ terms on the left} \\ X'(I_n - H_Z)X\hat{\beta}_1 &= X'(I_n - H_Z)Y && \text{Factor out } X' \text{ and recognize } I_n - H_Z \\ X'M_ZX\hat{\beta}_1 &= X'M_ZY && M_Z = I_n - H_Z \quad (13.3) \end{aligned}$$

The matrix $H_Z = Z(Z'Z)^{-1}Z'$ is the hat matrix from Chapter 9, built by the identical recipe, except that we fed it the controls instead of the full design matrix. It projects onto the column space of Z alone. So $I_n - H_Z$ is the residual maker for a regression on Z : hand it a vector and it hands back whatever part of that vector the controls could not explain. The controls have quietly turned themselves into a cleaning machine, and we did not ask them to.

Step 6: One cleaning becomes two.

Here is the step the whole theorem rests on. Equation (13.3) reads:

$$X' \underbrace{M_Z}_{\text{one cleaning}} X \hat{\beta}_1 = X' \underbrace{M_Z}_{\text{one cleaning}} Y.$$

There is a residual maker on each side, sitting in the middle, and neither X nor Y has been cleaned. Yet the theorem promised us *two* cleanings, one applied to X and one applied to Y , run as two separate regressions. Where is the second one supposed to come from? We cannot just conjure a matrix out of thin air.

We do not have to. Recall two facts about the residual maker from Chapter 9. It is **symmetric**, $M'_Z = M_Z$, and it is **idempotent**, $M_Z M_Z = M_Z$. Chain them together:

$$\begin{aligned} M_Z &= M_Z M_Z && \text{Idempotency: cleaning twice equals cleaning once} \\ &= M'_Z M_Z && \text{Symmetry: the first copy equals its own transpose} \end{aligned} \quad (13.4)$$

That is, *cleaning once is the same as cleaning twice*. Run X through the machine and every trace of Z is gone. Run the result through a second time and the machine finds nothing left to remove, so it hands back what you gave it because there is nothing left to take out. This means we are allowed to write a second M_Z into (13.3) for free. It costs nothing, because the second one does nothing. And once there are two of them, symmetry lets us hand one to the left and one to the right:

$$\begin{aligned} X' M_Z Y &= X' M'_Z M_Z Y && \text{Insert a free second cleaning, Equation (13.4)} \\ &= (M_Z X)' (M_Z Y) && \text{Regroup: } A' B' = (BA)', \text{ so } X' M'_Z = (M_Z X)' \\ &= r' v && \text{Definitions: } r = M_Z X, \ v = M_Z Y \end{aligned}$$

and running the identical argument on the left hand side, with X in place of Y ,

$$X' M_Z X = (M_Z X)' (M_Z X) = r' r.$$

We never decided to partial Z out of Y . We never chose to run two regressions instead of one. We took the single multiple regression, wrote down its normal equations, eliminated the nuisance coefficients, and the second residual calculation fell out of the algebra on its own. The lone M_Z in the middle of (13.3) was always secretly two of them, because an operation that repeats itself for free can be split in half and dealt out to whatever sits on either side. They were never two steps. They were one step, written twice.

Step 7: Solve for $\hat{\beta}_1$.

Substituting both results back into (13.3) and dividing:

$$\hat{\beta}_1 = \frac{X' M_Z Y}{X' M_Z X} = \frac{r' v}{r' r}.$$

Both of those are 1×1 , which is to say ordinary numbers, so dividing is legal.³ $\square$

13.2.1 Is That Really a Simple Regression?

We have proved $\hat{\beta}_1 = r' v / r' r$, and the statement of the theorem claimed this is the slope from a simple regression of v on r . Those are not obviously the same sentence.

Both r and v are $n \times 1$ columns, so $r' v$ and $r' r$ are just sums:

$$\hat{\beta}_1 = \frac{r' v}{r' r} = \frac{\sum_{i=1}^n r_i v_i}{\sum_{i=1}^n r_i^2}.$$

Now compare that with the simple regression slope we derived in Chapter 7, Equation (7.9):

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

These match only if $\bar{r} = 0$ and $\bar{v} = 0$, so that subtracting the means changes nothing. They are, and the reason is one we have known since Chapter 7.

Remember that r is the residual vector from a regression of X on Z . The normal equations of *that* regression force the residuals to be orthogonal to every column of Z . And Z contains a column of ones, because we absorbed the intercept into the controls. Orthogonality to the ones column says

$$\mathbf{1}' r = \sum_{i=1}^n r_i = 0 \quad \implies \quad \bar{r} = 0,$$

which is the same fact as $\sum_i \hat{u}_i = 0$ from simple regression, wearing different clothes. Any regression that includes an intercept produces residuals that sum to zero. The identical argument applies to v , so $\bar{v} = 0$ as well.

Substituting $\bar{r} = \bar{v} = 0$ into the Chapter 7 formula:

³ This is the one place we lean on X being a single column. If you want the coefficients on several variables of interest at once, the same proof goes through with X an $n \times p$ matrix and gives $\hat{\beta}_1 = (X' M_Z X)^{-1} X' M_Z Y$, with a matrix inverse where the division used to be.

$$\frac{\sum_i (r_i - \bar{r})(v_i - \bar{v})}{\sum_i (r_i - \bar{r})^2} = \frac{\sum_i (r_i - 0)(v_i - 0)}{\sum_i (r_i - 0)^2} = \frac{\sum_i r_i v_i}{\sum_i r_i^2} = \frac{r'v}{r'r}.$$

Not similar. Not proportional. The same expression, character for character. Step three of the FWL procedure is the simple regression from the very first chapter, run on residuals.

One practical crumb. Since r and v both have mean zero, the fitted line in that last regression passes through the origin, so its intercept comes out to exactly zero. Include an intercept in step three or leave it out. The slope will not move.

13.2.2 Do We Even Need Step Two?

Step one partials Z out of X and gives us r . Step two does the same to Y and gives us v . What happens if we skip step two and regress the raw Y on r ?

Split Y into the part the controls explain and the part they do not, which is precisely what the hat matrix and the residual maker do:

$$Y = \underbrace{H_Z Y}_{\text{what } Z \text{ explains}} + \underbrace{M_Z Y}_{= v}.$$

Now hit both sides with r' :

$$\begin{aligned} r'Y &= r'H_Z Y + r'v && \text{Distribute } r' \text{ across the split} \\ &= 0 + r'v && r \text{ is orthogonal to every column of } Z \\ &= r'v \end{aligned}$$

The first term dies because $H_Z Y$ is, by construction, some linear combination of the columns of Z , and r was built to be perpendicular to all of them. So $r'Y = r'v$, exactly, and therefore

$$\frac{r'Y}{r'r} = \frac{r'v}{r'r} = \hat{\beta}_1.$$

Step two is optional. For the point estimate, at least. Regress the raw Y on r and you get the same coefficient to the last decimal. It also explains *why* the two steps can be separated in the first place: once r has been made perpendicular to everything the controls can build, the part of Y that the controls explain is already invisible to it. Step two removes something r was never going to see.

So why bother with step two? Two reasons, and neither one is the coefficient. The first is the picture: plotting v against r centers both axes and shows the partial relationship honestly, which is the added variable plot of Section 13.7. The second is the standard error, which depends on the residual variance, and that is a topic we return to

in Chapter 14. Step two does not buy accuracy. It buys precision, and it buys a picture worth looking at.

13.2.3 Cashing the Promise

Enough algebra. We opened this chapter by claiming we could reproduce the OBP coefficient of 19.20 from Chapter 8 without running a multiple regression. Here is the whole thing: regress OBP on a constant and SLG, keep the residuals; regress runs per game on a constant and SLG, keep those residuals; take the ratio of the two sums.

Quantity	Value
$\bar{r}$ (mean of the OBP residuals)	-8.15×10^{-17}
$\bar{v}$ (mean of the R/G residuals)	1.22×10^{-15}
$r'v$	0.028513
$r'r$	0.001485
$r'v/r'r$ (the FWL slope)	19.197913
OBP coefficient, full regression	19.197913

Three things to notice. The means of r and v come back as numbers around 10^{-16} , which is a computer's way of writing zero: this is the $\bar{r} = \bar{v} = 0$ result from the previous subsection, rounded off by sixteen digits of floating point arithmetic. Run this yourself and you will see -8.15×10^{-17} rather than a clean 0. That is what exact zero looks like after a few thousand additions and subtractions in binary. The ratio $r'v/r'r$ is built from two sums that have nothing obviously to do with slugging percentage, and yet slugging percentage is fully accounted for, because it was removed from both ingredients before the sums were taken. And the FWL slope agrees with the full multiple regression to every digit displayed, because these are not two approximations of the same quantity. They are the same number, computed two ways.

Frisch and Waugh's point, translated out of 1933 time series language and into ours: partialing SLG out by hand first is not an alternative to controlling for SLG. It *is* controlling for SLG.

13.3 The Intuition: Partialing Out

Let's translate the algebra into a picture.

When you run a multiple regression of Y on X and Z , OLS is trying to isolate the part of X 's variation that is "its own" and not shared with Z . Think of it this way: